

SPA611

Mid-Semester Examination

Solutions

Section A: All questions carry 1 mark

1. Casseopeia A, a radio source at declination of $58^{\circ}48'54''$, when observed from a latitude of 58° North will:

(a) Never sets (b) Rises and sets (c) Never rises

Solution: The minimum elevation θ_{min} , of a source of declination δ from a latitude ϕ is:

$$\theta_{min} = \phi + \delta - 90^{\circ} \quad (1)$$

For the given declination and latitude, θ_{min} is $\sim 26^{\circ}$. Since the minimum elevation is > 0 , the source *never sets*.

2. A radio telescope has an SEFD ~ 1000 Jy and has a receiver with a bandwidth of 1 MHz connected to it. A source of 1 Jy can be detected with 5σ by this telescope with 1 second averaging.

(a) True (b) False

Solution: We will use the radiometer equation here:

$$\Delta S_{min} = \frac{SEFD}{\sqrt{\beta\tau}} \quad (2)$$

which gives the noise floor of the telescope (1σ). Substituting 1 MHz and 1 s in the SEFD formula, we find $\Delta S_{min} \sim 1$ Jy. Therefore, for the given bandwidth and integration time, we cannot detect a 1 Jy source at a 5σ significance.

3. An electron is spiralling around the earth's magnetic field. If the magnetic field of the earth was $20 \mu T$, the resulting EM radiation will be at:

(a) 560 kHz (b) 10 MHz (c) 50 Hz (d) 1420 MHz

Solution: An electron entering a magnetic field will go in a helical path around it, and will emit cyclotron emission. The frequency of cyclotron emission is $\frac{eB}{2\pi m_e}$, which when substituting the constant's values, turns out to be 560 kHz.

4. A telescope with an alt-az mount gives the same values of measured polarization for different pointings.

(a) True (b) False

Solution: For an alt-az mount, the plane of polarization will rotate wrt the sky plane. Due to this, the polarization that we measure will be modulated because of the parallactic angle.

5. Which of these spectra looks like that from a thermal source in radio frequencies?

Solution: d, according to Rayleigh Jeans's Law.

Section B: All questions carry 3 marks

1. The plasma dispersion equation is given using the following relation:

$$\omega^2 = \omega_p^2 + c^2 k^2 \quad (3)$$

where ω is the EM-wave frequency, while ω_p is the plasma frequency. Using this, find out what happens to an EM wave whose frequency is less than and greater than the plasma frequency? The earth's ionosphere has an electron density $\sim 10^5$ electrons per cm^3 . What will happen to a wave at 10 MHz aimed at the ionosphere?

Solution: Re-arranging the equation, we will get :

$$k = \frac{\omega}{c} \cdot \sqrt{1 - \frac{\omega_p^2}{\omega^2}} \quad (4)$$

When $\omega > \omega_p$, the value under root is real and > 0 , which implies that waves will be able to propagate, while $\omega < \omega_p$ implies k is imaginary. This implies that the wave will not propagate, and will be an evanescent wave.

Applying the plasma frequency formula, $\omega_p = \sqrt{\frac{n_e e^2}{m_e \epsilon}}$ and using the constants provided above, we will find $f_p \sim 3$ MHz. This implies the wave of 10 MHz will be able to penetrate the ionosphere and propagate beyond – in other words, the ionosphere will be ‘transparent’.

2. At optical wavelengths, the atmospheric fluctuations impose an angular resolution limit of $\sim 1''$. However, radio frequencies are not affected by these. If we require an angular resolution $\sim 1''$ at a frequency of 1 GHz, what would be the size of such a telescope? Is this physically possible to construct a telescope of this size? If this telescope (100% efficiency) was connected to a receiver of a 100 K noise temperature, what will be the flux of a radio source that will double the receiver temperature?

Solution: The angular resolution of a telescope is given by the diffraction limit:

$$\theta \approx \frac{\lambda}{D}$$

For a frequency $f = 1$ GHz,

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{10^9} = 0.3 \text{ m}$$

The required resolution is

$$\theta = 1'' = \frac{1}{206265} \text{ rad}$$

Thus, the required diameter is

$$D = \frac{\lambda}{\theta} = \frac{0.3}{1/206265} \approx 6.19 \times 10^4 \text{ m} \approx 62 \text{ km}$$

A single-dish telescope of diameter ~ 62 km is not physically feasible to construct. In practice, such resolutions are achieved using interferometric arrays (e.g., VLBI), which synthesize large apertures. The antenna temperature contribution from a source is

$$T_A = \frac{A_e S}{2k}$$

To double the receiver temperature ($T_{\text{rec}} = 100$ K), we require

$$T_A = 100 \text{ K}$$

For 100% efficiency, the effective area is

$$A_e = \frac{\pi D^2}{4} = \frac{\pi (6.19 \times 10^4)^2}{4} \approx 3.01 \times 10^9 \text{ m}^2$$

Thus, the required flux density is

$$S = \frac{2kT_A}{A_e} = \frac{2 \times (1.38 \times 10^{-23}) \times 100}{3.01 \times 10^9} \approx 9.17 \times 10^{-31} \text{ W m}^{-2} \text{ Hz}^{-1}$$

In Jansky units ($1 \text{ Jy} = 10^{-26} \text{ W m}^{-2} \text{ Hz}^{-1}$):

$$S \approx 9.17 \times 10^{-5} \text{ Jy} \approx 0.09 \text{ mJy}.$$

This is the SEFD of this huge dish!

Section C: The following question carries 4 marks:

- Two isotropic antennas are kept at a separation of d . What will be the power pattern that this would project on a “screen” at ∞ . How different will this be from the case where there is a ‘filled’ aperture extending from $-d/2$ to $+d/2$?

Solution:

Since we say at a distance of ∞ , we want to measure the angular/spatial distribution of power in the far field. We know that the far-field pattern is proportional to the Fourier transform of the aperture illumination function $A(x)$:

$$E(\theta) \propto \int A(x) e^{-ikx \sin \theta} dx$$

Let $u = \frac{\sin \theta}{\lambda}$, so that

$$E(u) \propto \int A(x) e^{-i2\pi ux} dx$$

Case 1: Two isotropic antennas separated by d The aperture distribution is two delta functions:

$$A(x) = \delta\left(x - \frac{d}{2}\right) + \delta\left(x + \frac{d}{2}\right)$$

Taking the Fourier transform:

$$E(u) \propto e^{-i2\pi u(d/2)} + e^{i2\pi u(d/2)} = 2 \cos(\pi ud)$$

Thus, the power pattern is

$$I(u) \propto |E(u)|^2 = 4 \cos^2(\pi ud)$$

In angular form:

$$I(\theta) \propto \cos^2\left(\frac{\pi d}{\lambda} \sin \theta\right)$$

Case 2: Filled aperture from $-d/2$ to $+d/2$ The aperture is a rectangular function:

$$A(x) = \Pi\left(\frac{x}{d}\right)$$

Its Fourier transform is:

$$E(u) \propto d \operatorname{sinc}(\pi u d)$$

where

$$\operatorname{sinc}(x) = \frac{\sin x}{x}$$

Thus, the power pattern is

$$I(u) \propto d^2 \operatorname{sinc}^2(\pi u d)$$

or

$$I(\theta) \propto \operatorname{sinc}^2\left(\frac{\pi d}{\lambda} \sin \theta\right)$$

Comparison:

- The two-element case produces a **pure fringe pattern**:

$$\cos^2\left(\frac{\pi d}{\lambda} \sin \theta\right)$$

with no envelope.

- The filled aperture produces a **diffraction pattern**:

$$\operatorname{sinc}^2\left(\frac{\pi d}{\lambda} \sin \theta\right)$$

with a central lobe and decaying sidelobes.

- The fringe spacing in the two-element case is $\sim \lambda/d$, which matches the main lobe width of the filled aperture.
- Physically, the two-point aperture samples only two spatial points, resulting in no suppression of sidelobes, while the filled aperture integrates continuously, and the destructive interference between different points on the aperture produces sidelobe decay.

Section D: All questions carry 5 marks

1. You have designed a new optical component that can potentially cause changes to polarization of light. You would like to test this newly designed “polarizer”. You have a light source, which can emit vertical and horizontally polarized light, and a “standard” vertical polarizer, and also a detector, that measures light’s intensity.
 - (a) How can you use the above components to test your “polarizer”? Explain with a diagram.
 - (b) You notice that the detector measures the same intensity level irrespective of the input polarization. What do you infer?
 - (c) Now, you rotate your vertical polarizer a full 360° and find that the intensity remains constant with respect to rotation angle. What could be the device you designed?

(a) i. **Experimental method and setup:**

We probe the polarization transformation by placing the unknown device between a known input state and an analyzer:

Source (H/V) $\longrightarrow$ Unknown Device $\longrightarrow$ V-Polarizer $\longrightarrow$ Detector

Procedure:

- Illuminate the device with horizontally and vertically polarized light.
- Measure transmitted intensity through a vertical analyzer.
- Then rotate the analyzer and record intensity as a function of angle.

ii. **Observation: Same intensity for H and V inputs**

If the detected intensity is the same for both horizontal and vertical inputs, this implies:

- The device does not preferentially attenuate one linear polarization over the other.
- Instead, it alters the *phase* between orthogonal components.

This is characteristic of a **wave plate** (a birefringent element), not a polarizer.

iii. **Observation: Intensity is constant as analyzer is rotated**

If the intensity remains constant as the analyzer is rotated through 360° , then the output light must be **circularly polarized**.

This is because:

- Circular polarization has equal projections onto all linear polarization directions.
- Hence, a rotating linear analyzer transmits constant intensity.

Conclusion:

The device converts linear polarization (H or V) into circular polarization. This occurs when:

- The device introduces a phase shift of $\pi/2$ between orthogonal components, and
- The input polarization is at 45° to its principal axes.

Therefore, the device is a **quarter-wave plate**.

2. You have a cavity which is divided into partitioned into two. On one side, you have an absorber surrounding the antenna from all sides, and on the other, you have a resistor of $50\ \Omega$. There is a filter between the partition that allows no em radiation, but power generated at the antenna can be transferred to the resistor.

- (a) What will be the power measured across the resistor?
- (b) If the resistor is connected to a receiver that is out of the cavity, and assuming that the receiver adds an excess power, P_n , due to its noise, what will be the total power measured at the output of the receiver?
- (c) Assuming you fill the cavity with warm air at a temperature T_{hot} , and liquid nitrogen at T_{cold} , and make your power measurements after giving sufficient time for the system to thermally settle down, how can you estimate the temperature of the receiver from this ?

Assume $\frac{h\nu}{kT} \ll 1$, and the measurements are done with identical bandwidth settings.

(a) i. **Power across the resistor:**

The absorber surrounding the antenna behaves as a blackbody at temperature T . Under the Rayleigh–Jeans limit ($\frac{h\nu}{kT} \ll 1$), the available noise power from a matched source over bandwidth $\Delta\nu$ is

$$P = kT\Delta\nu$$

Since the antenna is matched to the 50Ω resistor and the filter allows power transfer (but no EM leakage), the resistor measures the thermal noise power:

$$P = kT\Delta\nu$$

ii. **Power at receiver output:**

If the resistor is connected to a receiver that contributes a noise power P_n , the total measured power becomes:

$$P_{\text{tot}} = kT\Delta\nu + P_n$$

The receiver noise can be expressed as an equivalent noise temperature T_{rec} , where

$$P_n = kT_{\text{rec}}\Delta\nu$$

so that

$$P_{\text{tot}} = k(T + T_{\text{rec}})\Delta\nu$$

iii. **Estimating receiver temperature:**

Measure output power for two known physical temperatures:

$$P_{\text{hot}} = k(T_{\text{hot}} + T_{\text{rec}})\Delta\nu$$

$$P_{\text{cold}} = k(T_{\text{cold}} + T_{\text{rec}})\Delta\nu$$

Define the Y-factor:

$$Y = \frac{P_{\text{hot}}}{P_{\text{cold}}} = \frac{T_{\text{hot}} + T_{\text{rec}}}{T_{\text{cold}} + T_{\text{rec}}}$$

Solving for T_{rec} :

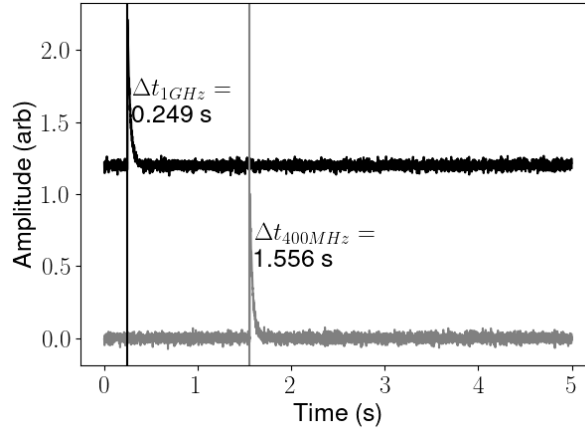
$$Y(T_{\text{cold}} + T_{\text{rec}}) = T_{\text{hot}} + T_{\text{rec}}$$

$$YT_{\text{cold}} + YT_{\text{rec}} = T_{\text{hot}} + T_{\text{rec}}$$

$$T_{\text{rec}}(Y - 1) = T_{\text{hot}} - YT_{\text{cold}}$$

$$T_{\text{rec}} = \frac{T_{\text{hot}} - YT_{\text{cold}}}{Y - 1}$$

3. Pulsars are neutron stars, and are known to have strong magnetic fields, and emit radiation out of its magnetic poles. We observe its radiation when its beam axis crosses our line of sight. You have a receiver working at two frequencies: 400 MHz and 1 GHz. The combined chart-recorder output of both these receivers show a



pattern as follows, and peak times of these pulses have been marked for you. The time taken by a pulsar signal to reach the observer at a frequency f is given using the formula [in mks units]:

$$t(f) = L/c + \frac{e^2}{8\pi^2\epsilon_0 m_e c} \cdot \frac{DM}{f^2} \quad (5)$$

where $DM = \int_0^L n_e(s) ds$.

- Find the DM of the pulsar observed here in units of parsecs per cm^3 .
- The heliosphere and the ionosphere also have electrons which might contribute to the line of sight electron density. The ‘thickness’ of the heliosphere is $\sim 10^{-3}$ pc, and that of the ionosphere is 10^{-11} pc. The electron densities in the heliosphere and the ionosphere is $\sim 10^{-2}$ and 10^5 cm^{-3} . The remaining path that the signal takes is through the interstellar medium with an electron density of 0.03 cm^{-3} . Find the approximate distance to the pulsar using this info.
- Comment on which region contributes the most to the DM, and why?

1 pc $\approx 3.0857 \times 10^{16}$ m.

- The dispersion delay between two frequencies is:

$$\Delta t = t(f_{\text{low}}) - t(f_{\text{high}}) = K DM \left(\frac{1}{f_{\text{low}}^2} - \frac{1}{f_{\text{high}}^2} \right)$$

where, in practical units,

$$\Delta t \text{ (s)} \approx 4.15 \times 10^{-3} DM (f_{\text{low}}^{-2} - f_{\text{high}}^{-2})$$

with f in GHz and DM in pc cm^{-3} .

From the plot:

$$\Delta t = 1.556 - 0.249 = 1.307 \text{ s}$$

Using $f_{\text{low}} = 0.4 \text{ GHz}$ and $f_{\text{high}} = 1 \text{ GHz}$:

$$\Delta t = 4.15 \times 10^{-3} DM (0.4^{-2} - 1^{-2})$$

$$= 4.15 \times 10^{-3} DM (6.25 - 1) = 4.15 \times 10^{-3} \times 5.25 DM$$

$$\Delta t \approx 0.0218 DM$$

Thus,

$$DM = \frac{1.307}{0.0218} \approx 60 \text{ pc cm}^{-3}$$

$$\boxed{DM \approx 60 \text{ pc cm}^{-3}}$$

(b) Now estimate contributions:

Heliosphere:

$$DM_{\text{helio}} = (10^{-3} \text{ pc})(10^{-2} \text{ cm}^{-3}) = 10^{-5} \text{ pc cm}^{-3}$$

Ionosphere:

$$DM_{\text{iono}} = (10^{-11} \text{ pc})(10^5 \text{ cm}^{-3}) = 10^{-6} \text{ pc cm}^{-3}$$

Both are negligible compared to total $DM \sim 60$.

Thus, ISM dominates:

$$DM \approx n_e L$$

$$60 \approx 0.03 \times L \Rightarrow L \approx \frac{60}{0.03} = 2000 \text{ pc}$$

$$\boxed{L \approx 2 \times 10^3 \text{ pc}}$$

(c) **Dominant contribution:** Interstellar medium (ISM)

Although the ionosphere has a very high electron density and the heliosphere has moderate density, their physical thicknesses are extremely small. The ISM, despite its low density ($\sim 0.03 \text{ cm}^{-3}$), extends over kiloparsec scales.

Since

$$DM = \int n_e ds$$

the long path length through the ISM makes it the dominant contributor.

$$\boxed{\text{ISM dominates due to its very large path length}}$$